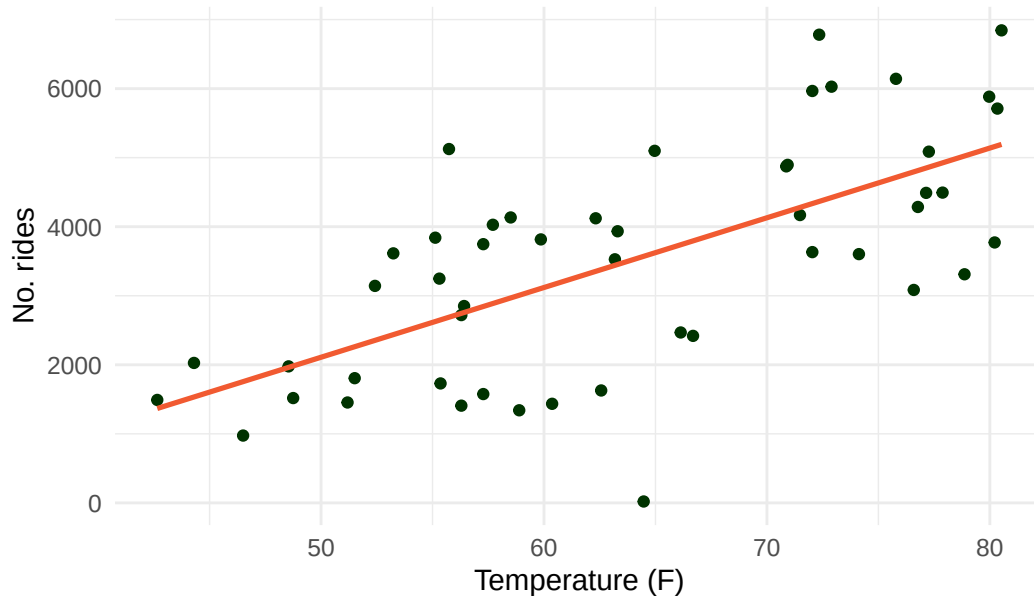


Day 05: Evaluating SLR models

Name: _____

Data for today

We return to the Capital Bikeshare data, this time working with a small random sample of 50 days so that patterns are easier to see by eye.



term	estimate	std. error	t	p-value
Intercept	-2940.400	1072.078	-2.74	0.009
temp_actual	100.995	16.567	6.10	<0.001

Two definitions of a "good" model

1. The model is appropriate, so our inferences (p-values, CIs) are _____
2. The x variable does a good job _____ the y variable

Theoretical model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ where $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$

Fitted model: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, so that $y_i = \hat{y}_i + \hat{\varepsilon}_i$

Conditions for inference: LINE

For our p-values and confidence intervals to be trustworthy, the theoretical model above has to be a reasonable description of the data. Check:

- **Linear:** _____
- **Independent and identically distributed errors:** _____
- **Normally distributed errors:** _____
- **Equal variance of the errors:** _____

Note

These conditions are about the *errors*, not about the raw x or y variables themselves.

Residuals

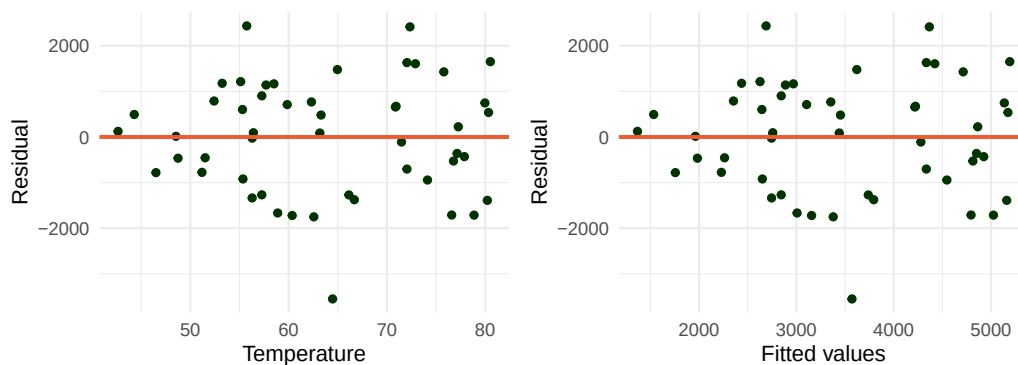
Definition: $e_i = \hat{\varepsilon}_i = y_i - \hat{y}_i$

Properties:

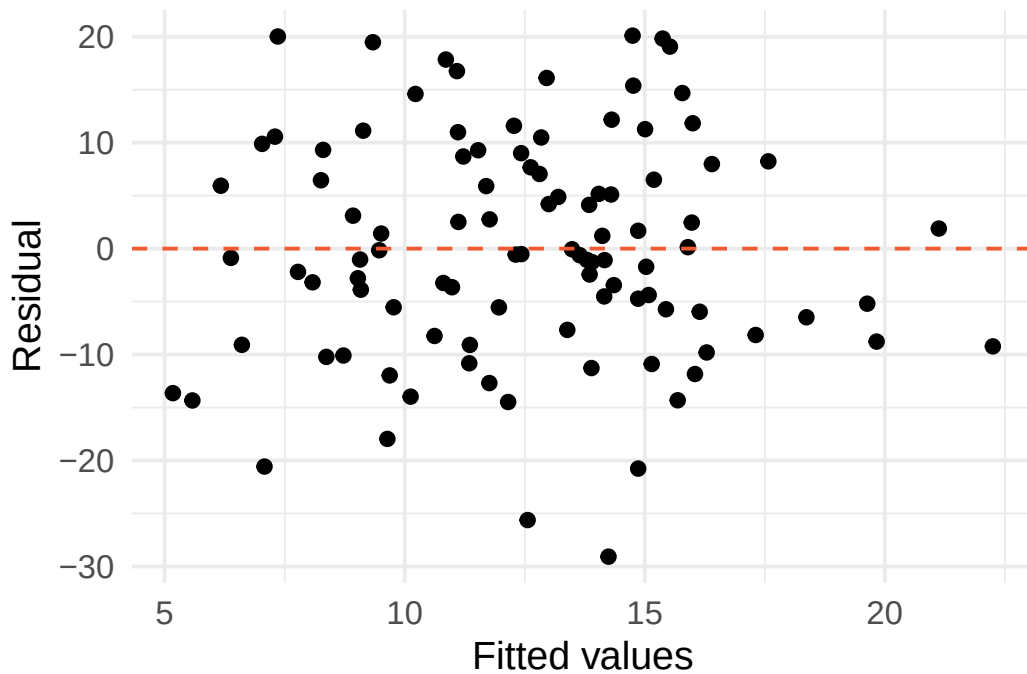
- sum to zero $\implies$ mean is ____
- **uncorrelated** with x and $\hat{y}$
- (should be) normally distributed
- $SD(e_i) = \hat{\sigma} \sqrt{1 - \frac{1}{n} - \frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$

Residual plots

For SLR, plotting residuals against x or against the fitted values $\hat{y}$ shows the same shape (since $\hat{y}$ is a linear function of x) – either is a valid way to check linearity and equal variance.



A “good” residual plot shows no pattern: residuals bounce randomly around zero with roughly constant spread across the range of fitted values.



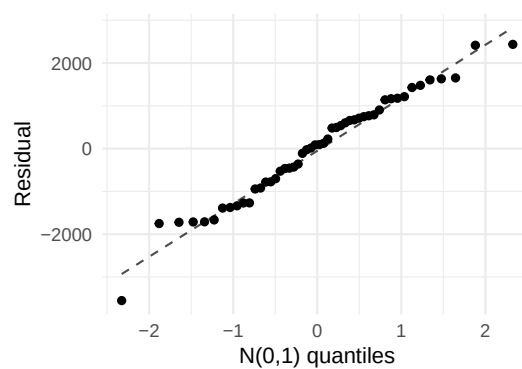
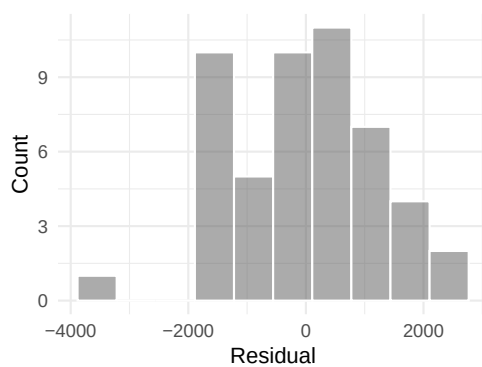
💡 Group task

On the whiteboards, sketch a plot of y vs. x and the corresponding residual plot that would indicate a violation of

1. the linearity condition
2. the equal variance condition

Assessing normality

Use a histogram or normal Q-Q plot of the residuals. A “good” Q-Q plot hugs the diagonal reference line closely.

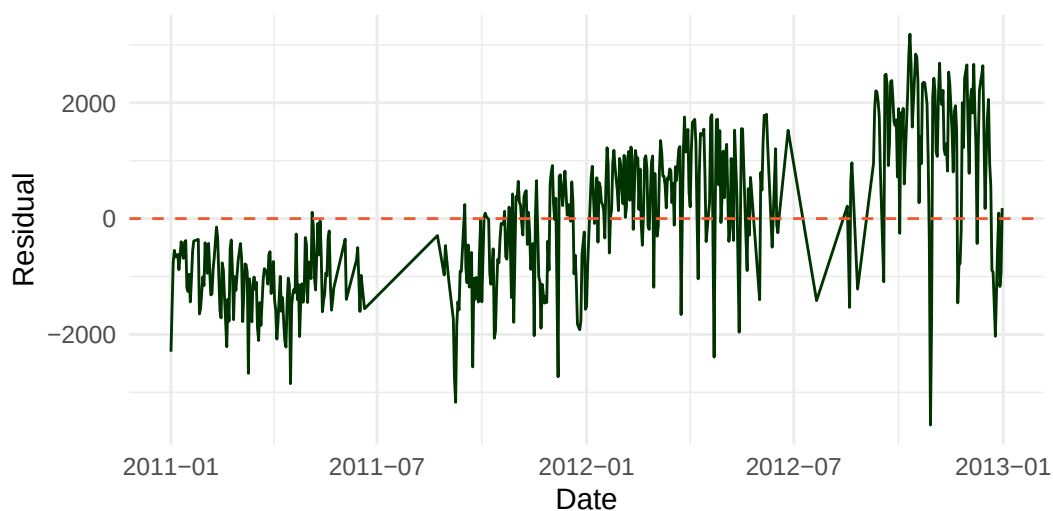


i Note

Normality is **harder to judge at small sample sizes** – even true random draws from a normal distribution can look lumpy or have a wandering Q-Q plot when n is small. Don't over-interpret minor wiggles with only 30-50 points.

Assessing independence

Plot residuals against whatever variable might induce dependence (time, location, subject, batch, ...). Using the *full* bikes dataset ordered by date reveals a clear pattern – a strong hint that day-to-day ridership is **not** independent once we condition only on temperature:



What happens if the conditions aren't met?

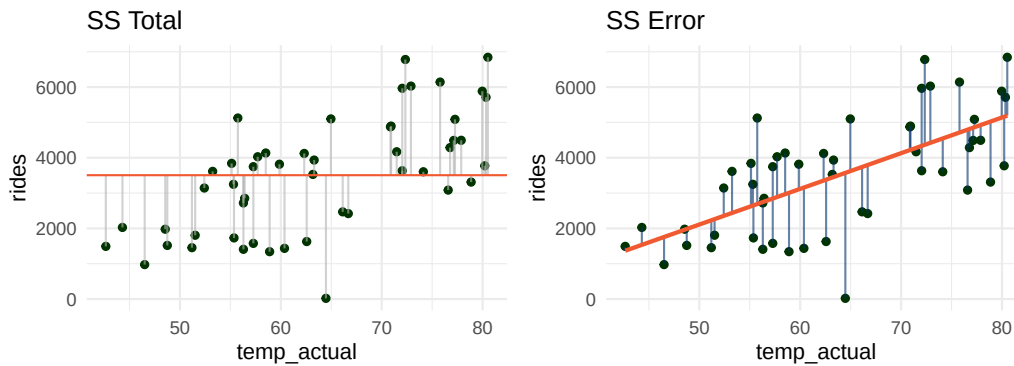
- **Linearity:** if nonlinear, everything breaks!
- **Independence:** estimates are still unbiased, but the SEs are typically too **small** (we have less information than n independent points suggests)
- **Normality:** estimates and SEs are still fine, but **prediction intervals** (which rely on individual points being normal around the mean) are wrong
- **Equal variance:** estimates are still unbiased, but SEs are wrong – and we don't know in which direction

What do we do if assumptions are violated?

1. Change our assumptions (hard – needs more advanced methods)
2. Transform y , x , or both
3. Change the type of inference (e.g. bootstrap)

Is x a “good” predictor of y ?

Partitioning variability



$$\underbrace{\sum (y_i - \bar{y})^2}_{SSTO} = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum e_i^2}_{SSE}$$

total variability = variability of $\hat{Y}$'s + variability of residuals (_____)

$$R^2 = \frac{SSR}{SSTO} = \frac{\text{variability of } \hat{Y}}{\text{variability of } Y}$$

i R^2

R^2 gives the proportion of total variability in Y explained by the SLR model.

For our small bike sample, $R^2 = 0.436$ – about 43.6% of the variability in rides is explained by temperature.

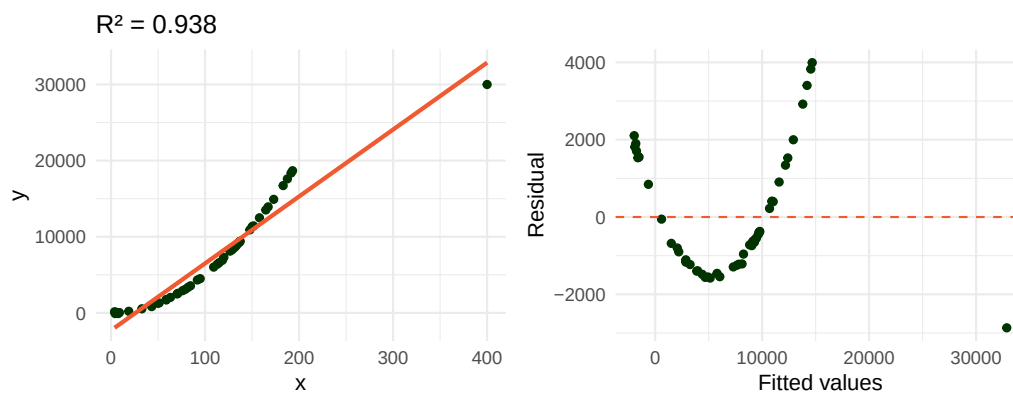
Properties of R^2

- Between ____ and ____
- Big when σ is small relative to the overall variability of y (i.e. a big proportion of variability explained by the model)
- For SLR, $R^2 = r^2$ (where r is the correlation coefficient between x and y)

Quick notation reference

Symbol	Meaning
r	sample correlation coefficient
ρ	population correlation coefficient
R^2	coefficient of determination
p	a population proportion
$\hat{p}$	a sample proportion
p-value	$P(\text{test statistic} \mid H_0)$
$P(A)$	probability of event A

R^2 dangers: high R^2 doesn't mean ``good fit''



A single high-leverage point drove R^2 up to 0.938 even though the residual plot shows an obvious curve – the linear model is a bad fit! **Always check the residual plot, not just R^2 .**

Your turn

Would you trust this model just based on its R^2 ? What does the residual plot tell you that R^2 alone can't?